

# Joel Hempel

## Machine Learning and Full-Stack Development

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## Summary Statement

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Driven software and hardware enthusiast with a strong track record in full-stack development and machine learning. Known for exceeding expectations and swiftly mastering new technologies.

## Technologies

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**Languages:** Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS

**Frameworks & Libraries:** PyTorch, React, Next.js, FastAPI

**Web/Databases:** Supabase, SQL, Tailwind, PHP

**Tools:** Git, Docker, AWS, Google Cloud, JUnit, Google Colab

**ML:** Neural Networks, Computer Vision, Language Models

## Experience

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### Software Sub-Team

*UBC Subbots*

**Vancouver, BC**

*09/2024 - Present*

- Developing trajectory generation (navigation) algorithms for AUV.
- Integrating IMU with Arduino and sensors for improved underwater navigation.
- Collaborating with other sub-teams

### SWE Intern (ML & Full-Stack)

*Metriized Consulting Inc.*

**Vancouver, BC**

*09/2023 - 08/2024*

- Led full-stack application development deployed within 12 months.
- Enhanced YOLO-based object detection, introducing key features that improved performance.
- Established E2E/integration testing, trained and tuned neural networks.
- Conducted academic research implementations, data manipulation, and synthetic data generation.
- Prepared investor presentations; thoroughly documented a 20k-line project.

### Undergraduate Research Assistant

*Systopia Lab*

**Vancouver, BC**

*05/2023 - 12/2023*

- Researched ML/deep learning to detect spring onset with satellite/climate data.
- Optimized data preprocessing for stronger predictive models.
- Co-authored and presented a paper at NeurIPS 2023: [\(Link\)](#)

### IT Intern

*Industrial Light & Magic (Lucasfilm)*

**Vancouver, BC**

*05/2022 - 08/2022*

- Maintained and troubleshooted hardware/software in a production environment.
- Achieved highest ticket completion rate on the team.

## Education

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### BSc, Computer Science

*University of British Columbia*

**Vancouver, Canada**

*2022–2025*

- **GPA:** 4.00/4.33

### Transfer Credits

*Langara College*

**Vancouver, Canada**

*2022–2025*

- **GPA:** 4.17/4.33

### Awards:

- Ernest E. Livesey Memorial Prize in Mathematics.
- LSU Faculty of Science Award (highest cumulative GPA).